Krishna Balaji

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EDUCATION

University of Illinois Urbana-Champaign **Aug 2026 - May 2028**
Master of Computer Science *Champaign, IL*
- **Coursework:** Distributed Systems, Database Systems, Applied Machine Learning

Arizona State University **Aug 2022 - May 2026**
B.S. in Computer Science (Honors), Minor in Business, GPA 4.00/4.00 *Tempe, AZ*
- **Coursework:** Data Structures & Algorithms, Operating Systems, Software Engineering, Object-Oriented Programming, Artificial Intelligence, Linear Algebra, Probability & Statistics, Discrete Mathematics

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript/TypeScript, SQL, HTML/CSS
Frameworks & APIs: React, Next.js, Node.js, Express, FastAPI, Flask, Playwright, LangGraph, REST APIs
Databases: PostgreSQL, MongoDB, Redis, pgvector
Cloud & DevOps: Docker, Linux, AWS (EC2, S3, Lambda), GCP, Git, GitHub Actions, CI/CD
ML/Data: Pandas, NumPy, scikit-learn, XGBoost, LightGBM, Polars, Parquet, PuLP, LangChain

EXPERIENCE

Software Engineering Intern **Sep 2025 - Apr 2026**
Circle.ooo *Tempe, AZ*
- Implemented 41 REST API endpoints across 9 Node.js/Express route modules for a TypeScript sponsorship platform, supporting core marketplace, payment, and real-time messaging workflows.
- Designed a sponsorship recommendation engine scoring compatibility between hosts and sponsors across 6 weighted factors including industry overlap, budget fit, and audience type, surfacing ranked matches across both dashboards.
- Engineered sponsorship workflows across 9 relational database tables for proposal management, deadline enforcement, messaging, and Stripe checkout integration.

Research Software Engineer Intern **Sep 2025 - Apr 2026**
Arizona State University *Tempe, AZ*
- Engineered a Python ML pipeline processing 106,881 player gameweek observations across four Premier League seasons, using XGBoost and LightGBM with walk-forward validation and achieving 0.70 mean Spearman correlation.
- Designed and validated a rank-based scoring signal across 11M+ historical Fantasy Premier League team selection records and four seasons of out-of-sample data, outperforming the crowd baseline by 0.96 points per gameweek across all evaluation windows.
- Built a Mixed Integer Linear Programming optimizer in PuLP for team selection under budget and position constraints, ranking in the top 0.9% on a blind 2024-25 holdout.

Machine Learning Engineer Intern **May 2025 - Aug 2025**
Quantiphi *Mumbai, India*
- Developed Python ETL pipelines with Pydantic schema validation to process 50,000+ insurance claims per month, reducing manual review time by 80% through automated field extraction and anomaly flagging.
- Built document extraction microservices using Gemini 1.5 Pro, structured outputs, and confidence thresholding, improving extraction accuracy from 72% to 90% across 10,000+ claims.
- Deployed document extraction microservices on GCP with automated validation and error handling, helping reduce claim approval cycle time by 25%.

PROJECTS

BrowserProof: Reliability Testing for AI Browser Agents **Aug 2026 - Sep 2026**
- Built a reliability testing platform for browser agents with 10 browser and network fault types, including ambiguous commits, stale DOM state, session expiry, and lost responses, verifying that agents recover without causing incorrect external actions.
- Traced agent actions through network requests and backend commits, detecting 90% of seeded failures across 36 controlled scenarios and converting failures into reproducible CI regression tests.

Glassbox SRE: Autonomous Incident Response Agent **May 2026 - Jul 2026**
- Built an incident response pipeline with FastAPI and Redis that processes Prometheus alerts through parallel LangGraph workflows for commit correlation, runbook retrieval, and impact analysis, storing results in PostgreSQL and delivering Slack incident briefs.
- Designed a replayable 15-scenario benchmark with same-service distractors, achieving 86.7% top-1 commit identification accuracy versus 13.3% for deterministic heuristics, with 100% runbook and impact classification accuracy.

ATLAS: High-Performance Analytics Engine **Feb 2026 - Mar 2026**
- Built an analytics service with async FastAPI endpoints and Redis caching for factor analytics and regime detection across 4,000+ stock tickers over 10-year windows, achieving sub-200ms P99 latency versus a 1.4s Pandas baseline.
- Built Polars and Parquet ingestion pipelines with fault handling and market-based caching across 15M+ price data points, cutting backtest query time by 80% and supporting rolling cross-sectional regressions.